

ML4SCI GSoC 2026 — Task 2c

Agentic AI for HEP Analyses (HEPTAPOD CMS Layout)

Repository `KenWuqianghao/heptapod` (branch `gsoc26-cms-restructure`)

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1 Objective

Per the ML4SCI evaluation brief: clone HEPTAPOD, work on a **separate branch** in your fork, propose an **alternative folder structure** and **tool set** informed by repositories linked on the project page, and **do not open a pull request** to the upstream HEPTAPOD repository unless the program explicitly requests it.

2 Fork and branch

Development branch: `gsoc26-cms-restructure`. Example clone:

```
git clone -b gsoc26-cms-restructure \
  https://github.com/KenWuqianghao/heptapod.git
```

Upstream reference: <https://github.com/tonymenzo/heptapod>.

3 Proposed layout (`cms_agent/`)

A parallel tree separates CMS-oriented orchestration from the existing generic `tools/`, `llm/`, and `examples/` directories:

- `cms_agent/agents/` — planner contract and role boundaries.
- `cms_agent/workflows/` — example run cards and JSON schema for validated planner output (where applicable).
- `cms_agent/tools/adapters/` — thin adapters toward CMSSW, `coffea`, `awkward`, etc., preserving HEPTAPOD-friendly JSON outputs.

- `cms_agent/configs/` — model routing and tool allowlists (e.g. forbid arbitrary shell).
- `cms_agent/knowledge/` — curated snippets / RAG manifest placeholders.
- `cms_agent/eval/` — golden prompts and small fixtures (e.g. histogram contract checks).

Architecture narrative: `ARCHITECTURE_CMS_AGENT.md` at repository root.

4 Tool inventory

`cms_agent/TOOL_INVENTORY.md` maps each conceptual tool to dependencies cited in HEPTA-POD documentation (Orchestral AI runtime, MCP bridges, PDG/INSPIRE tools in-repo, CMSSW/coffea-class stacks for CMS-specific steps) and records failure modes.

5 Minimal working adapter

`cms_agent/tools/adapters/cms_histograms.py` implements a **NumPy-only** leading- p_T histogram JSON contract so the scaffold can be exercised without CMSSW or `coffea` installed.

6 Compliance

Work remains on the contributor fork branch; no merge to upstream was required for the evaluation checklist.

Separation from Task 2b

Task 2b (jet image super-resolution GAN) is maintained in a separate repository: https://github.com/KenWuqianghao/CMS_E2E.

Report artifact

This PDF is built from `docs/GSoC26_Task2c_Report.tex` in this fork.